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Microfluidic Flow Cytometer for Quantifying Photobleaching of Fluorescent Proteins in Cells

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Supporting Information

ABSTRACT: Traditional flow cytometers are capable of rapid cellular assays on the basis of fluorescence intensity and light scatter. Microfluidic flow cytometers have largely followed the same path of technological development as their traditional counterparts; however, the significantly smaller transport distance and resulting lower cell speeds in microchannels provides for the opportunity to detect novel spectroscopic signatures based on multiple, nontemporally coincident excitation beams. Here, we characterize the design and operation of a cytometer with a three-beam, probe/bleach/probe geometry, employing HeLa suspension cells expressing fluorescent proteins. The data collection rate exceeds 20 cells/s under a range of beam intensities (5 kW to 179 kW/cm^2). The measured percent photobleaching (ratio of fluorescence intensities excited by the first and third beams: S_{beam3}/S_{beam1}) partially resolves a mixture of four red fluorescent proteins in mixed samples. Photokinetic simulations are presented and demonstrate that the percent photobleaching reflects a combination of the reversible and irreversible photobleaching kinetics. By introducing a photobleaching optical signature, which complements traditional fluorescence intensity-based detection, this method adds another dimension to multichannel fluorescence cytometry and provides a means for flow-cytometry-based screening of directed libraries of fluorescent protein photobleaching.

Traditional flow cytometers employ light-scattering and fluorescence-based detection to assess spectral diversity,[1−3] cell size,[4] fluorescence brightness,[5−7] fluorescence lifetime,[8,9] and analyte concentration[10] on individual cells flowing through one or more tightly focused excitation beams at speeds of ∼1−10 m/s. In this operating regime, the time window for optical excitation and detection is approximately a few microseconds per beam and hundreds of microseconds between beams. In contrast, the short transport dimensions and confining properties of microfluidic channels enable highly stable flows at cell speeds of 10^{-6} to 10^{-3} m/s. We exploited these properties to develop the ability to screen with optical or photophysical properties that are manifested at longer timescales (tens of milliseconds or slower) by implementing multipoint fluorescence excitation measurements in a microfluidic flow cell. We specifically investigate the probing of photobleaching in flow. Although it is likely to be ubiquitous in flow cytometry, few studies have investigated photobleaching in detail, and there are no reports of fluorophore screening or sorting based on photobleaching. Previous reports by van den Engh et al. and Doornbos et al. focused on understanding photobleaching and photon saturation in DNA stains, fluorescein conjugates, phycoerythrin, and allophycocyanin, via pulse shape and power-dependence measurements, primarily with the goal of optimizing the magnitude of fluorescence signals.[11,12] The excitation conditions in those studies accessed time windows of 2 μs to 2 ms, at excitation intensities of 5−3200 kW/cm^2. Here, we report the design of a cytometer for assessing photobleaching of genetically encodable fluorescent proteins, at excitation intensities comparable to those used for confocal imaging and single molecule spectroscopy (10−100 kW/cm^2).

Since the advent of green fluorescent protein (GFP), genetically encodable fluorescent proteins (FPs)[13] in a diversity of excitation and emission wavelengths have found widespread use in molecular and cell biology due to the ability to fuse them to a protein of interest and target them to specific subcellular structures.[14] Despite these advantages, FPs exhibit complex excited-state dynamics which limit their photon output. In these fluorophores, irreversible photobleaching, which refers to photodestruction of the chromophore, usually occurs in the presence of reversible photobleaching, which involves transient conversion to a nonfluorescent or dimly fluorescent state. Depending on the FP, reversible photobleaching has been attributed to triplet state conversion,[15] excited state proton-transfer,[16] or photoinduced isomerization of the chromophore and nearby side chains.[17−19] Subsequent ground-state recovery occurs in tens of microseconds to minutes[20,21] and may depend on the chromophore environment of the FP.[21] In ensemble measurements, reversible photobleaching is manifested as an initial fast decay of fluorescence that recovers when the excitation light is turned off.[14,21−23] The magnitudes and time-constants of both reversible and irreversible photobleaching depend on the fluorophore, excitation intensity, and excitation wavelength.[14,21−23] Although these factors make the composite photobleaching process tricky to quantify, there is a clear

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potential for using it in flow-cytometric screening for the development of new fluorescent proteins. This approach would be significantly faster than microplate or colony-based screening. Due to the longer time scale photophysics in FPs compared to small molecule fluorophores, a cytometry-based screening system will require that a correspondingly longer time-window be accessible to the measurement.

In recent work, we reported the use of millisecond pulse sequences to dissect the photobleaching process of FPs in individual HeLa cells.[23] Here, to measure photobleaching of cells in flow, we implemented a design that quantifies photobleaching on the millisecond time scale, independent of fluorophore concentration, fluorescence quantum yield or extinction coefficient. We employ three spatially separated beams: a low-intensity probe beam (5 kW/cm^2) to measure initial fluorescence, followed by a high-intensity bleach beam (5−179 kW/cm^2) to initiate photodestruction of the fluorophores, followed by a second low-intensity probe beam, of equal intensity to the first, to assess the extent of photobleaching. This approach simplifies data acquisition compared to direct measurement of a time-resolved fluorescence decay because ratios of peak signal intensities are easier to fit and define than the multiexponential decays which characterize the photobleaching process.

Here, we combine microfluidics and spectroscopy techniques in a flow cytometer for measuring the combined effect of reversible and irreversible photobleaching at a rate of >20 cells/s. To our knowledge, this capability has not previously been reported. We present calculations guiding the optical design of a multibeam cytometer describing how to optimize measurement precision and alignment tolerance in a simple 2D hydrofocusing geometry. The technique is demonstrated on four different red fluorescent proteins (RFPs), under a variety of excitation conditions, which uncovered a diverse range of reversible and irreversible photobleaching on the millisecond time scale. Lastly, we present kinetic simulations to examine the effects of reversible and irreversible photobleaching rates on the capability of our method to discriminate populations.

■ EXPERIMENTAL SECTION

The cytometer consists of four general components: (1) a microfluidic hydrofocusing chip and manifold which holds the sample and drives fluid flow, (2) an inverted microscope which serves as an optical platform for the cytometer, (3) optics to shape and align the three beams into the microscope, (4) the data acquisition and processing electronics.

Design Considerations. One of the design goals was to ensure that the distribution of measured fluorescence intensities for a population accurately reflects cellular RFP expression heterogeneity rather than instrument resolution. Even for a single-beam fluorescence measurement of each cell in flow, the excitation intensity and therefore the observed emission signal is strongly dependent on the trajectory of each cell as it traverses through the 3-D laser focus in the microchannel. Although, in principle, 3-D hydrofocusing geometries would more precisely define these cellular trajectories, we show that simpler-to-fabricate 2D hydrofocusing devices can provide sufficient precision for properly designed multibeam excitation/detection geometries. In 2-D hydrofocusing, cells flowing past a cross-junction with two channels of sheath flow at higher pressures are laterally confined by the flow to dimensions significantly narrower than the channel width (Figure 1a).[24] Nevertheless, there will always be lateral and axial variation (relative to the optical axis of the microscope objective: see

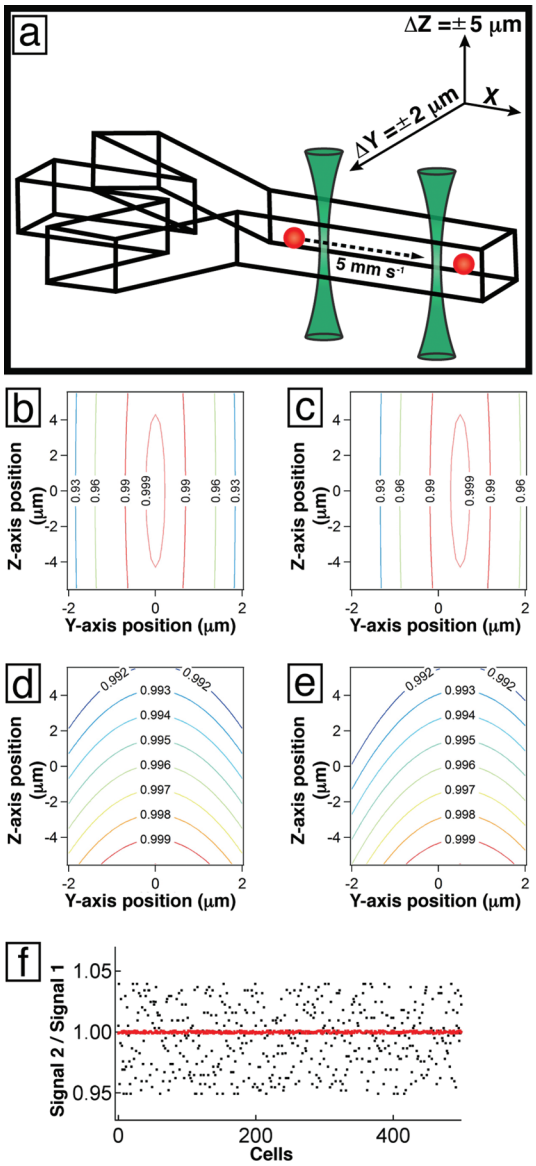

Figure 1. (a) Schematic of hydrodynamically focused cells traveling through two circularly shaped Gaussian laser beams where ΔZ and ΔY refer to the cell axial and lateral displacement from the center of the channel. (b−e) Contour plots showing peak fluorescence signal as a function of the cell displacement (ΔZ and ΔY) as it flows through circularly (b, c) and elliptically (d, e) shaped Gaussian beams. Parts b and d represent the fluorescence signal if the two beams are perfectly centered in the channel, whereas parts c and e represent the fluorescence signal if the second beam is displaced 0.5 μm along the y-axis relative to the center of the channel. (f) For the misaligned beams, a scatter plot of the ratio of peak fluorescence signals (I_1/I_2) from 500 cells randomly displaced from the center of the channel (within ΔY = ±2 μm and ΔZ= ±5.5 μm) shows that the signal variance from elliptically shaped beams (red) is 35-fold smaller than that from circularly shaped beams (black).

Figure 1) in trajectories from cell to cell. We quantify the effects of this variation on the fluorescence signal, along with the effects of slight misalignment of the two probe beams relative to the flow axis of the microfluidic. We minimized the alignment sensitivity by introducing a cylindrical lens (f = 150 mm) positioned ∼190 mm before the objective lens, to shape the Gaussian beam into an elliptical profile in the microchannel, in which the focused beam size perpendicular to the flow

direction (y-axis) is much larger than along the flow direction (x-axis).

To quantify the impact of cell transit variation on the fluorescence signal, we first calculated the astigmatic transformation of a Gaussian beam through a cylindrical lens and objective optical system with a number of experimental constraints. We assumed that each cell was 14 μm in diameter (the mean cell size observed for HeLa-S cells on a wide-field microscope), had a uniform RFP concentration (i.e., an RFP-containing sphere), and traveled between two 4 μm diameter (fwhm) laser beams (Figure 1a,b). This beam size matched our measurements of the focused probe beam sizes produced with a 20×, 0.45 NA air-objective (Figure 2). The observed fluorescence signal,

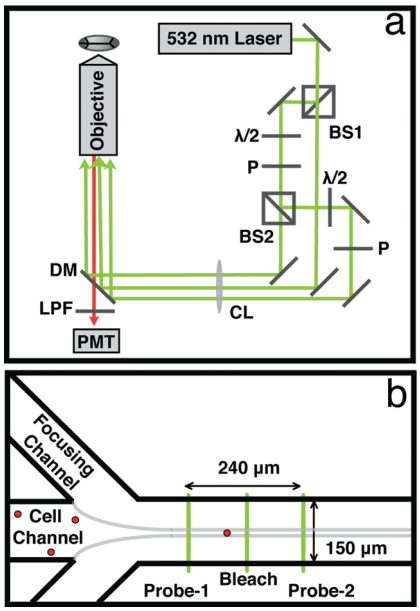

Figure 2. Three-beam microfluidic cytometer experimental design. (a) Schematic of the optical setup. Relevant components include the following: 20× 0.45 NA air immersion objective (obj); 532 nm dichroic mirror (DM); 545 nm long-pass filter (LPF); red-enhanced photomultiplier tube (PMT); 150 mm focal length, cylindrical lens (CL, placed 19 cm from back aperture of objective); half-wave plate ($\lambda/2$); Glan-Thompson polarizer (P); 70:30 beam splitter (BS1); 50:50 beam splitter (BS2). (b) Schematic of the microfluidic channel geometry at the interrogation region. Cells were hydrodynamically focused to a width of 15 μm before encountering the elliptical bleach and probe beams (fwhm 3 × 75 μm).

S (eq 1), is the convolution of the RFP density, $s(x,y,z)$, and the three-dimensional Gaussian intensity profile at the laser focus, $B(x,y,z)$:

$$S(x_o, y_o, z_o) = \iiint_{x,y,z} s(x - x_o, y - y_o, z - z_o) \times B(x, y, z)\, dx\, dy\, dz \quad (1)$$

Details of the analytical solution for a symmetrical Gaussian beam are provided in Supporting Information. To obtain an analytical expression for the asymmetric Gaussian beam shaped by the cylindrical lens, we used the transfer matrix system analogous to the ABCD matrix method for geometric optics[25] and calculated the evolution of the beam in the x and y directions, independently.

Hence, the elliptical Gaussian beam may be written as

$$B[x,y,z] = \frac{2P}{\pi w_x[z]\, w_y[z]} \times \exp\left[-\left(\frac{2x^2}{w_x^2[z]} + \frac{2y^2}{w_y^2[z]}\right)\right] \quad (2)$$

where P is the total power of the beam, and $w_x[z]$ and $w_y[z]$ are, respectively, the x-axis and y-axis beam radius where the intensity drops to $1/e^2$ of its peak value. The evolution of the Gaussian beam along the propagation direction, defined as the z-axis here, is

$$w_i[z] = w_0\left(1 + \frac{z^2}{z_{Ri}^2}\right) \quad (3)$$

where i denotes x or y direction, and $z_{Ri} = 2\pi w_{0i}^2/\lambda$ is known as the Rayleigh range. At $z = z_{Ri}$ the beam radius is $\sqrt{2}$ times larger than its waist value w_0, and the beam area doubles.

The calculated beam radius at the sample, which is positioned at the x-axis waist position, is $w_x[z]|_{z=z0} = 2$ μm and $w_y[z]|_{z=z0} = 56$ μm, which is consistent with the measured beam radius of $w_x = 3$ μm and $w_y = 67$ μm. The distribution of peak signal intensities from a cell traversing a spherically and cylindrically focused beam centered on the hydrodynamic flow are shown in Figure 1b and 1d, respectively. These contour plots reveal how the observed fluorescence signal varies as a function of the cell axial and lateral position with respect to the center of the channel ($\Delta Z = \Delta Y = 0$). Figure 1c and 1e are similar but correspond to the case in which the beam is displaced laterally ($\Delta Y = 0.5$ μm) with respect to the center of the channel. This displacement represents the precision of experimental alignment. Because of the relatively low NA of the optical system, the signal intensity is insensitive to cell axial positioning (±5.5 μm is the maximum range for a 14 μm diameter cell in a channel of 25 μm height) for both spherical and cylindrical focusing. However, in the lateral direction, a comparison of spherical vs cylindrical beam shaping reveals very different sensitivity to cell position and beam alignment. In particular, if the beams are misaligned or the cell drifts by $\Delta Y = 0.5$ μm, the difference in the peak fluorescence intensity of the second beam relative to the first beam is significantly smaller with the cylindrical focus (Figure 1d,e). Consequently, we consider cells transiting only along the X-axis between the two probe beams at randomly chosen axial and lateral positions in the range $\Delta Y = \pm 2$ μm and $\Delta Z = \pm 5.5$ μm. If the two beams are perfectly aligned, then for both cases, the ratio of fluorescence intensities (beam1/beam2) = 1. However, for the misaligned geometry, this signal ratio depends on the cell position in the channel. Figure 1g shows that cylindrically focused beams yield a 35-fold lower dispersion in the signal intensities compared to the spherically focused beams. This result indicates that pairs of cylindrically focused beams will lead to substantially smaller variability in fluorescence measurements.

Microfluidics and Optical Layout. Microfluidic devices were built by anodically bonding a 25 μm thick 2 in. diameter silicon wafer to a 1.7 mm thick glass-slide. Silicon was etched down to the glass in the pattern of the channels using standard photolithography and plasma etching techniques.[26] This method results in optically transparent channels of 25 μm height × 150 μm width × 1 mm length for the central interrogation channel. Sample ports of 1 mm diameter were drilled in a second, identical glass slide before bonding to the silicon.

The microfluidic was compression fit with "O" rings against a manifold constructed from polytetrafluoroethylene (PTFE; to minimize nonspecific adsorption of cells) with 200 μL sample reservoirs (Supporting Information Figure S-1). The microfluidic device and combined manifold assembly were mounted onto the stage of a commercial inverted microscope. Flow was driven using three closed-loop air-pressure controllers connected by PTFE tubing to the sample ports. By independently varying the pressures on all three inlets, the hydrodynamic focal width was kept constant at 15 μm as measured by imaging the fluorescence from a dye flowing in only the center cell channel.[24] The cell speed in the interrogation region was varied from 1 to 15 mm/s to control exposure time to the bleach beam. The speed was calculated from measurements of cell transit times between probe beams 1 and 2 using fluorescence signals, and measurements of spatial separation of the beams (Figure 2; typically 240 ± 3 μm). The cell speed distribution typically had a standard deviation of 1%. The flow was visualized with a CMOS camera, and wide-field transillumination was provided by a home-built condenser.

The three-beam geometry consisted of two equal intensity probe beams measuring the peak fluorescence from a cell before and after a higher-intensity photobleaching beam. To implement this experimental geometry, a 2 W 532 nm continuous wave laser was split into three beams by a series of beam splitters (30:70 and 50:50) and waveplate-polarizer pairs, thereby allowing independent control of each beam's excitation intensity. After splitting, all beams were shaped by a cylindrical lens (150 mm focal length), directed into the microscope, reflected from a 532 nm dichroic mirror, and focused inside of the microfluidic channels by a 20×, 0.45 NA air-objective (Figure 2a). Shaping the beams with the cylindrical lens results in elliptical beams (75 μm length × 3.5 μm width, fwhm, as measured by imaging light scattered from the sample focal plane onto a CMOS camera) that stretch the entire width of the hydrofocus. The beams were distributed over a 240 μm distance with the bleach beam located midway between the two probe beams (Figure 2b). The probe beam intensity was 5 kW/cm^2 and the bleach beam was 170 kW/cm^2 (calculated from the fwhm of the beam dimensions and laser power measured at the sample plane (±5%)). The probe beams were matched in intensity before each experiment. Fluorescence was collected through the same objective and separated from excitation light by the 532 nm dichroic mirror and a 545 nm long-pass filter. The emission was detected by a red-sensitive photomultiplier tube (PMT, Hamamatsu, R9880U-20) on the primary imaging port of the microscope. At this port, the fluorescence signals from the three beams are spatially resolved, which allowed for placement of a mask at the focal plane which blocks the photobleaching beam. A lens is used to refocus fluorescence from the two probe beams onto the PMT.

Data Acquisition and Processing. The PMT photocurrent was processed by a custom-built AC-coupled transimpedance operational amplifier, which improves the signal-to-noise ratio by removing high and low frequency noise components outside the band-pass of 160 mHz to 106 kHz. The resulting voltage levels were digitized at 250 kHz with a PC-based data acquisition board (16 bit ADC) and custom software (LabView, National Instruments). After fitting each peak to a Gaussian, the peak fluorescence signals for the first and second probe beams (S_{beam1} and S_{beam3}) were recorded. Typical fluorophore transit times through each beam varied from 0.2 to 3.5 ms depending on the cell velocity (1−15 mm/s) and neglecting fluorophore diffusion within the cell.

Sample Preparation. HeLa suspension (HeLa-S, mean diameter = 14.4 μm, CV = 21.8%, as measured from images taken on a widefield microscope) cells were maintained in spinner flasks at 37 °C in a 5% CO_2 atmosphere using spinner-modified Dulbecco's Modified Eagles Medium, 10% fetal bovine serum, and 1% penicillin−streptomycin. HeLa-S cells were virally transduced according to manufacturer's protocols with an FP either under a constitutive cytomegalovirus promoter (TagRFP, and TagRFP-T in pCLNCX) or an inducible tet-responsive promoter (mCherry, mOrange2, pCL-TRE). Briefly, virus was generated by transfecting the appropriate combination of DNA (pCLNCX-FP, pCLTRE-FP, pCL-Ampho, pCL-TetOn, pVSV-G) into HEK293FT cells. After two days, the virus-containing supernatant was collected, passed through a 0.45 μm cellulose acetate filter, and added to HeLa suspension cells with 12 μg/mL Polybrene and, if appropriate, expression was induced with 1 μg/mL doxycycline. To establish the cell lines, the fluorescent population was enriched once by fluorescence-activated cell sorting (FACS) with a Dako Cytomation Mo-Flo cell sorter, to eliminate cells that were not infected with virus (further details provided in Supporting Information). To avoid biasing the fluorescent populations evaluated in our experiments, no further enrichments by FACS were performed.

Aliquots of cells were concentrated via swinging-bucket centrifugation at 1000 rpm for 5 min. To prevent clumping and settling within the microfluidic reservoirs, cell pellets were resuspended in a density-matched medium using a commercially available density-matching solution and HEPES-buffered Hank's balanced salt solution (HHBSS), pH 7.4 solution, supplemented with 1% bovine serum albumin. Experiments involving beads utilized 6 μm diameter fluorescently labeled beads from an Invitrogen LinearFlow Deep Red Flow Cytometry Intensity Calibration Kit, suspended in a density-matched 20% (v/v) glycerol in water solution. The microfluidic channels were passivated with a 1% solution of bovine serum albumin prior to each run. Cell suspensions were loaded into the center reservoir in 150 μL aliquots at a concentration of ~5 × 10^5 cells/mL. The side channels were filled with 150 μL aliquots of HHBSS, pH = 7.4, for the sheath flow.

RESULTS AND DISCUSSION

The measured quantity in all experiments described below is the %bleach, which is defined in terms of the measured peak fluorescence signal for the first and third beams (S_{beam1} and S_{beam3}, respectively), as

$$\%\text{bleach} = \left[1 - \left(\frac{S_{beam3}}{S_{beam1}}\right)_{\substack{\text{with}\\\text{bleach}\\\text{beam}}} \times \left(\frac{S_{beam3}}{S_{beam1}}\right)_{\substack{\text{no}\\\text{bleach}\\\text{beam}}}\right] \times 100\% \quad (4)$$

To correct for small differences in beam intensity, and lateral misalignments of probe beams, the %bleach is defined as the function of reference measurements taken in the absence of the bleach beam. Note that %bleach may be composed of a combination of reversible and irreversible photobleaching, as dictated by the excited-state dynamics of the fluorophores. To shed light on the molecular origins of the measured %bleach in

terms of the rate constants for reversible and irreversible photobleaching, we present and discuss numerical simulations of the signals in terms of a four-state model of RFP photophysics

Single-RFP Population Photobleaching. We first performed multibeam fluorescence measurements on fluorescently labeled beads to verify the precision of the measurement-matched predictions from the design considerations discussed above. Data from one probe beam yields a fluorescence intensity coefficient of variation (CV) ranging from 6% to 16%, depending on the fluorescence intensity of the bead type (higher intensity beads yielded lower CV, Supporting Information Figure S-2b). The CV value averaged over all three populations of beads was within 10% of the average value obtained on a Dako Cytomation Mo-Flo FACS (Supporting Information, page S-4). This variability is lower than many other 2D-focusing microfluidic cytometers (CV of 25−30%)[27,28] and comparable to 3D-focusing microfluidic cytometers (CV of 1−9%)[29,30] but remains larger than state-of-the-art flow cytometers (e.g., CV < 3%, BD FACSAria). For a two-probe beam measurement with a mixture of beads, a plot of S_{beam3} vs S_{beam1} was linear, with a coefficient of determination, R^2 = 0.99 with a 7% CV in the ratio of S_{beam3} vs S_{beam1} for greater than 3000 events (Supporting Information Figure S-2a).

Two-beam measurements (without a bleach beam) on HeLa-S cells expressing TagRFP-T were fit to a line with a coefficient of determination of 0.99 and an 11% CV in the ratio of S_{beam3} vs S_{beam1} (Figure 3). In principle, for two probe beams of identical intensity, and in the absence of photobleaching, we expect a slope of 1 for a plot of S_{beam3} vs S_{beam1}. In general, we observed slopes of slightly less than 1 for both the beads (0.93, CV = 7%, Supporting Information Figure S-2a) and TagRFP-T-expressing cells (0.94, CV = 11%) which were statistically the same by an unpaired t test (T = 1.42). Under these probe beam conditions, we expect photobleaching of the beads, and even the less photostable RFPs, to be negligible. For example, using photobleaching kinetics parameters measured for TagRFP-T at 532 nm in immobilized HeLa cells at similar intensities,[23] we estimate 0.4% photobleaching occurs. It seems likely that the nonunity slope occurs primarily due to a slight mismatch in the probe beam power transmitted through the objective, which we observe to be highly sensitive to alignment into the microscope. In our definition of %bleach, we account for this mismatch in probe beam intensities, to provide for a corrected measure of the bleaching magnitude. For TagRFP-T cells, with a bleach beam intensity of 170 kW/cm^2 and flow speed of 27.9 mm/s (exposure time of 125 μs), the slope decreases to 0.52 (Figure 3), which indicates a significant amount of bleaching (%bleach = 51%, from eq 4). The same performance for all measurements were reproduced with the beams intersecting the hydrofocused stream anywhere along the ∼1 mm length from immediately after the hydrofocus to the end of the interrogation channel.

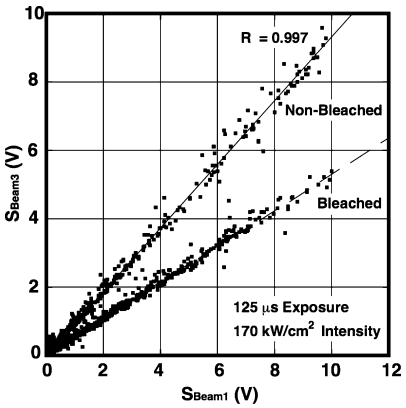

Figure 3. Single cell photobleaching of HeLa-S cells transduced with TagRFP-T interrogated with the two probe beams indicate that the cytometer's response is linear with respect to fluorescence intensity and the signals from each probe beam are highly correlated (CV = 11%, R^2 = 0.997, nonbleached slope = 0.94). Upon addition of the bleach beam, the slope decreases (bleached slope = 0.52), indicating that photobleaching occurs. Here, each point represents a measurement performed on a single cell.

Photobleaching of Fluorescent Protein Populations. To evaluate the ability to resolve subpopulations on the basis of photobleaching, three-beam measurements were performed on a mixture of HeLa-S cells expressing TagRFP,[31] TagRFP-T,[22] mOrange-2,[22] and mCherry.[32] Photophysical properties of these FPs are summarized in Supporting Information Table S-1. With a bleach beam intensity of 170 kW/cm^2 and flow speed of 12.9 mm/s (exposure time of 270 μs), the beam spacing resulted in an 8 ms average cell transit time between beams. This time scale permits complete recovery from higher-ordered excited states and dark states[23] (Supporting Information Table S-2). Under these conditions, four populations of cells were apparent in the plots of S_{beam3} vs S_{beam1} (Figure 4a). Each RFP-expressing cell population was identified by measurements on the individual cell types under identical flow and intensity conditions. A histogram of %bleach for the ∼1891 cells in this sample also reveals four subpopulations, corresponding to the four RFPs (Figure 4b). The rank order of average fluorescence intensities for the cell lines measured in the microfluidic cytometer agreed with those measured by FACS (TagRFP-T = TagRFP > mOrange2 > mCherry). The differences in fluorescence brightness for different RFP-expressing cell lines may result from differences in the relative absorption cross-section at 532 nm (Supporting Information Table S-1) or from differences in cellular RFP concentrations, which in turn may result from incomplete chromophore maturation and differences in the transcription promoter strength. As stated previously, the cells assayed in the cytometer were not prescreened or enriched for brightness; therefore, a large range (CV > 130%) of intensities were screened. Tuning the PMT gain to optimize detection of weakly fluorescent cells would permit improved resolution of the photobleaching response in these cells.[33] In Figure 4c, we plot the measured %bleach vs prebleach fluorescence intensity for the cell mixture. These data show a resolution of the mixture into four populations and demonstrate that the measured %bleach depends on the RFP but is largely independent of the fluorophore concentration (as given by the prebleach emission level). The ability of %bleach to resolve the mixture of four RFPs may be quantified by fitting the histogram (Figure 4b) to a sum of four Gaussians. The fit parameters (Figure 4 caption) reveal that the mean %bleach values for mOrange2, mCherry, and TagRFP were separated by at least 1σ, whereas the TagRFP-T population was separated from the others by at least 2σ. The percentages of cells that could be uniquely assigned to one population with at least 99.9% confidence were obtained by determining the confidence interval of the Gaussian fit for a given cell population which has less than 0.1% overlap from the

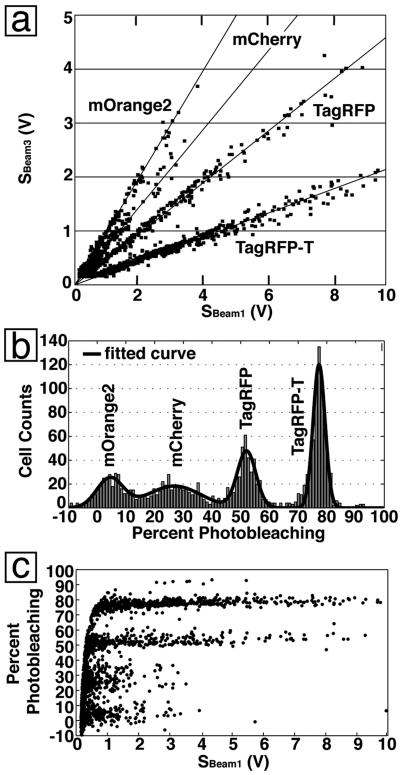

Figure 4. Resolving RFPs based on photobleaching in a microfluidic cytometer. (a) A mixture of cells expressing one of four RFPs was resolved on the basis of photobleaching. Each point represents an individually assayed cell and the slope of the S_{beam1} versus S_{beam3} plot yields the %bleach for each RFP. (b) The mean %bleach for each RFP-expressing cell line (upon measurement of 200−300 cells) was 4.4 (CV = 145%) %bleach for mOrange2, 26.8 (CV = 49%) %bleach for mCherry, 52.0 (CV = 9%) %bleach for TagRFP, and 77.3 (CV = 4%) %bleach for TagRFP-T as determined using a fitting program which fit a sum of four Gaussian functions to the histogrammed data. (c) Plot of %bleach vs prebleach intensity, showing the resolving power provided by bleaching measurements. The signal level corresponding to cellular autofluorescence is below the baseline of 27.8 mV.

Gaussian fits for the other cell populations. This confidence interval defines the percentage of cells in a population that can be assigned to a given Gaussian fit with 99.9% confidence. The percentages resolved by this criterion are as follows: 1% of the mOrange 2 cells were resolved from the mCherry cells, 43% of the mCherry cells from the mOrange 2 and TagRFP cells, 10% of the TagRFP cells from the mCherry and TagRFP-T cells, and 100% of the TagRFP-T cells from the others. More details on this resolution can be found in Supporting Information.

Photokinetic Simulations. To understand the connection between microfluidic photobleaching measurements and fluorophore photophysics, we performed numerical simulations based on a four-state system consisting of the ground state (S_0), the first excited state or bright state (S_1), and two dark or weakly fluorescent states (D_0 and D_1) (Figure 5a). Table 1 contains a summary of the reactions and kinetic parameters. Photobleaching was permitted out of S_1 and D_1. Recently we used this four-state model to describe the excited-state dynamics of RFPs in immobilized single cells exposed to a series of millisecond time scale excitation pulses and demonstrated that this model faithfully captured trends for reversible and irreversible photobleaching for a panel of 13 FPs.[23] A similar model has been used to examine reversible photobleaching (i.e., blinking) of GFP.[34] We modified our previous simulations by approximating the excitation profile as a sum of Gaussian pulses that replicate the durations and

Table 1. Summary of Reactions and Equations Used in the Photokinetic Model

process	reaction	reaction rate	rate constant value (s^{-1})
emission and nonradiative decay	$S_1 \rightarrow S_0$	$[S_1]\ k_{em}$	$k_{em} = 1 \times 10^9$
reversible photobleaching	$S_1 \rightarrow D_1$	$[S_1]\ k_{rev}$	k_{rev} = varied = 0 to 1×10^6
dark-state decay	$D_1 \rightarrow D_0$	$[D_1]\ k_{DSD}$	$k_{DSD} = 1 \times 10^9$
ground-state recovery	$D_0 \rightarrow S_0$	$[D_o]\ k_{GSR}$	$k_{GSR} = 1 \times 10^4$
S_1 bleach	$S_1 \rightarrow$ null	$[S_1]\ k_{S1bleach}$	$k_{S1bleach}$ = varied = 0 to 5×10^6
D_1 bleach	$D_1 \rightarrow$ null	$[D_1]\ k_{D1bleach}$	$k_{D1bleach}$ = varied = 0 to 5×10^{10}
excitation	$S_0 \rightarrow S_1$	$[S_0]\ k_{ex}$	k_{ex} = Gaussian

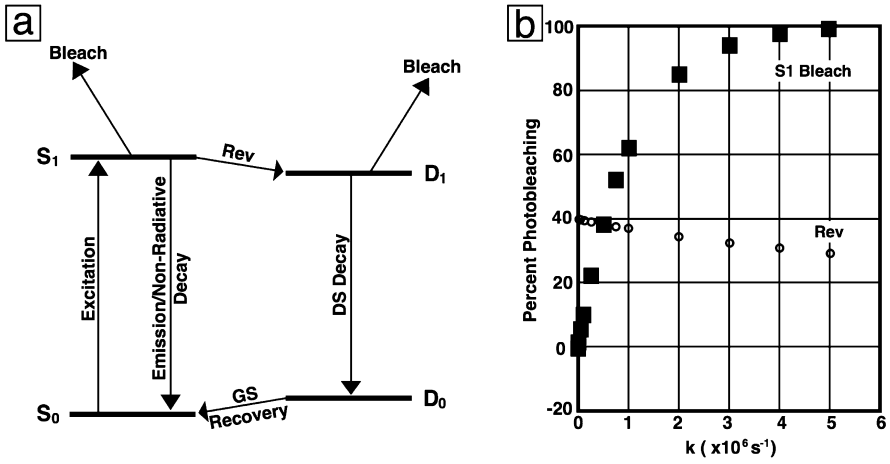

Figure 5. Kinetic modeling results. (a) Jablonski diagram depicting four-state model used for photokinetic modeling results. (b) Simulated percent photobleaching as a function of $k_{S1bleach}$ and k_{rev}. Increasing rates of irreversible photobleaching out of the first excited singlet state and reversible photobleaching have opposite effects on observed photobleaching.

excitation rates experienced by the fluorophores as they flow through the three beams in the cytometer.

Previous investigations of photodynamics in flow cytometry primarily focused on photobleaching and photon saturation,[11,12] which are the dominant processes operative at the $\sim 10^3$ kW/cm^2 intensities and microsecond time scales considered in those investigations. Photon saturation occurs when the average time between excitation−photon absorption approaches the time the fluorescent molecule spends in the excited state. We estimate that the average arrival time between excitation photon for our highest intensity beam (170 kW/cm^2) beam was 1.7 photons/μs. Because the excited-state lifetime of the RFPs are in the range of 2−3 ns,[23] photon saturation is negligible under the conditions employed here. We therefore focused on photobleaching and dark-state conversion processes.

The excitation rate was calculated using the measured, average-excitation intensity to find the peak-excitation intensity, which was then used to calculate the maximum rate of excitation for a representative RFP (TagRFP-T, which was chosen because its photophysical properties represent the median of the four proteins assayed) using its extinction coefficient at 532 nm (52000 M^{-1} cm^{-1}) and the Beer−Lambert law:

$$k_{\mathrm{ex}} = \frac{\sigma I \lambda}{\hbar c} = 2.303 \times 1000\left(\frac{\varepsilon I \lambda}{Nhc}\right) \tag{5}$$

where σ is the absorption cross-section, I is the light intensity, λ is the wavelength, h is Planck's constant ($\hbar = h/2\pi$), c is the speed of light, ε is the decadic molar extinction coefficient, and N is the number of molecules.

The fluorophores first experience an excitation rate corresponding to the first probe beam. The rate of excitation increases and then decreases in a Gaussian profile in time from zero up to the peak rate of excitation (8×10^4 s^{-1}) and then back to zero over the course of 0.54 ms. The excitation rate remains at zero for 20 ms (as mentioned previously, ground state recovery is complete after 8 ms) before experiencing the excitation of the bleach beam (maximum rate of 1.6×10^6 s^{-1}) and, last, the third probe beam. In accord with the calculations and measurements on the cylindrical beam shaping, the excitation profile was assumed to be constant in the direction perpendicular to the cell's travel. The fwhm of the laser spatial profile was transformed into time coordinates assuming an average cell velocity of 6 mm/s which is approximately the midpoint of the range of speeds used in these experiments. The peak of the time-dependent fluorescence profiles from the first and third excitation beams was then used to calculate the %bleach.

The values of the rate constants for each step in the four-state model were taken from our previous work (Table 1).[23] In particular, three parameters were varied individually and the magnitude of photobleaching was calculated for each simulation. First, the rate of bleaching out of the higher-energy dark-state (k_{D1bleach}) was varied while k_{rev} and k_{S1bleach} were held at 5×10^5 s^{-1}. A negligible increase in %bleach was observed for all but extremely large rate constants (1×10^{10} s^{-1}) indicating that, in this model, the dark state acts photoprotectively, i.e., the fluorophore does not bleach out of the dark state. Next, the rate of bleaching out of the first excited state (k_{S1bleach}) was increased from 0 to 5×10^6 s^{-1} while k_{rev} was held at 5×10^5 s^{-1} and k_{D1bleach} was held at 0. This perturbation resulted in an expected increase in %bleach because the increased rate of bleaching allowed the bleaching process to compete more successfully with the other S_1 depopulation pathways. Lastly, the rate of reversible photobleaching (k_{rev}) was increased over the same range while k_{S1bleach} was held at 5×10^5 s^{-1} and k_{D1bleach} was held at 0, leading to a decrease in % bleach. This trend shows that reversible and irreversible photobleaching are competing processes because an increase in the rate of either process leads to opposite impacts on the observed %bleach. Although the results of our model indicate that the k_{S1bleach} has an effect greater than that of k_{rev} on the observed values of %bleach, note that, in general, the rates of both processes are known to change with excitation intensity, pulsed vs continuous wave illumination, and excitation wavelength.[21−23]

For the four RFPs investigated here, the rates of reversible and irreversible photobleaching vary over 1−2 orders of magnitude across the range of excitation intensities characteristic of widefield and confocal microscopies (10 W/cm^2 to 1 kW/cm^2) (Supporting Information Table S-1).[22] Therefore, for completeness, these calculations were performed using a range of rate values (Table 1), covering both experimental and modeling estimates.[12,21,23,34−36] Our kinetic simulations indicate it is likely that the rates of both reversible and irreversible photobleaching are in the range from 1×10^5 to 1×10^6 s^{-1} because experimental and modeling results for the observed % bleach are in agreement in this range. Furthermore, these simulations predict that, for the current set of measurements, the magnitudes of reversible and irreversible photobleaching are anticorrelated (Figure 5b). The RFPs examined have rate constants that vary by 1 order of magnitude or less, and our modeling results indicate that, over this small range, the effects of reversible photobleaching significantly influence the observed extent of irreversible photobleaching.

The four proteins examined in this study represent closely related fluorophores. In particular, TagRFP and TagRFP-T differ by only one point mutation and have similar fluorescence spectra and quantum yields (Supporting Information Table S-1). However, as emphasized by the resolution of the populations in Figure 4, the FPs differ significantly in their photostabilities and propensities for reversible photobleaching. Our photokinetic simulations show that although the four cell populations are differentiated with our three-beam geometry, they are not resolved solely on the basis of reversible or irreversible photobleaching but rather by a combination of both processes. For this reason, additional resolving power will be necessary if the two processes are to be separated. For example, according to pulsed photobleaching experiments performed on stationary, individual cells (Supporting Information Figure S-3 and Tables S-1 and S-2), mCherry is less prone to both irreversible and reversible photobleaching than mOrange2. However, the significant contribution of reversible photobleaching for mOrange2 causes it to appear very similar to mCherry, if only %bleach is considered. Building on the multibeam geometry described here, other excitation schemes may be devised to separately measure the rates of both processes.

■ CONCLUSIONS

To our knowledge, this is the first cytometer designed to quantify photobleaching in mixed populations. This multibeam flow cytometer capitalizes on the spatiotemporal properties of the cells in microfluidic flow to measure photobleaching with a ratiometric approach, which inherently differs from previous experiments with one or two beams. We demonstrated the

capability to characterize the individual cells within a mixed population and note that our experiment resolves two RFPs (Tag-RFP[31] and Tag-RFP-T[22]) that cannot be resolved by previously available spectral signatures (e.g., fluorescence lifetime, excitation/emission spectra, absorption).

As revealed in these experiments and simulations, this three-beam pulse sequence probes both irreversible and reversible photobleaching. These two processes are highly interdependent and must be considered in tandem. To discriminate reversible from irreversible photobleaching, the excitation pulse sequence would need to operate on time scales that create a steady-state dark-state population. Generally, by controlling the cell velocity, excitation intensity, and dimensions of the beam and fluidic channel and by employing time-resolved fluorescence detection, it will be possible to implement specific probes of other photophysical dynamics, at high throughput. Furthermore, building on a recent suggestion in the literature,[37] the method reported here can be integrated with measurements of fluorescence lifetime[8,9] and microfluidic cell-sorting techniques[38,39] to enable the screening of genetic libraries of FPs on the basis of photostability and fluorescence quantum yield. This work, which is currently underway in our laboratory, will enable the development of a next generation of more photostable FPs.

■ ASSOCIATED CONTENT

Supporting Information

Additional information as noted in the text. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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